

DARPA Secure Handhelds on Assured Resilient networks at the tactical Edge (SHARE)

Secure data sharing at multiple security levels between coalitions across
mobile tactical networks

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Dr. Mary Schurgot, DARPA/STO

SHARE Overview

Program Goal: Demonstrate the secure exchange of information at multiple security levels over commercial and military networks using a heterogeneous mix of devices

Motivation: Named Data Networking (NDN) and increasingly powerful handhelds allow for efficient and reliable peer-to-peer communication at the tactical edge

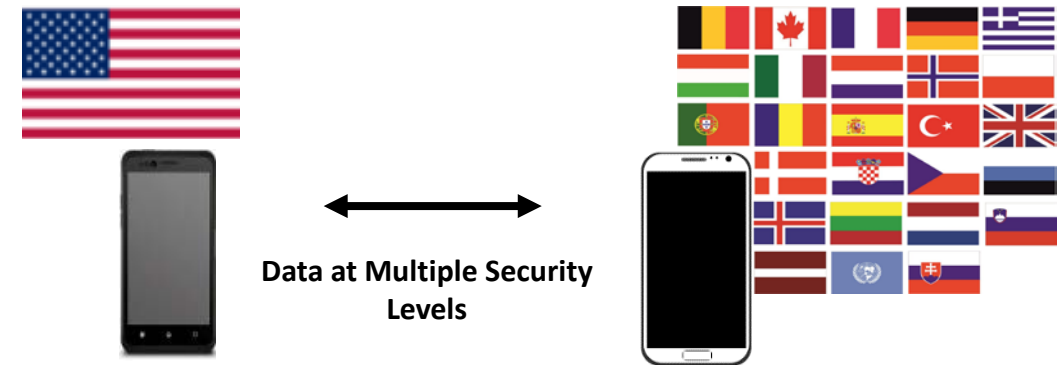
Performance & Technical Objectives

- Enable groups (disparate blue forces and coalition partners) to share data efficiently, securely, and reliably
- Provide distributed tactical data security management on trustworthy mobile devices
- Increase network security resiliency with built-in encryption and associated keys
- Automate security configuration management of sharing rules

Approaches & Challenges

- Distributed security based on secure mobile devices
- Resilient secure networks that are stable in challenging environments
- Software to rapidly configure security across the network

Tactical Information Sharing Using Mobile Handheld Devices

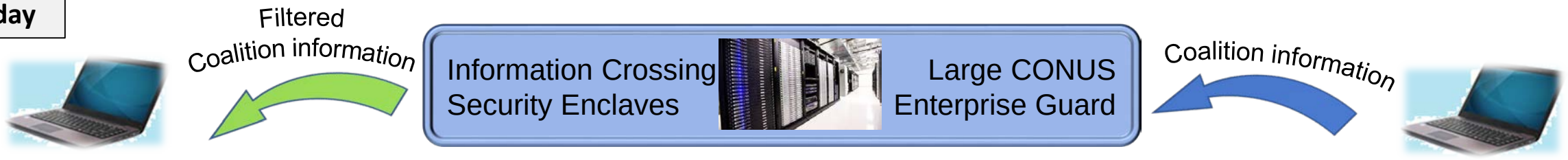


SHARE interfaces with the Tactical Assault Kit (TAK)

Secure data sharing at multiple security levels between coalitions across mobile tactical networks

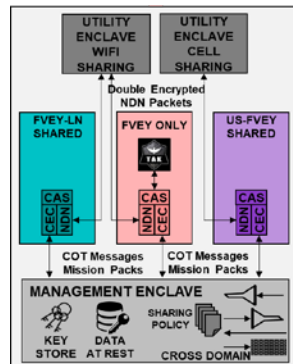
SHARE Technical Approach

Today

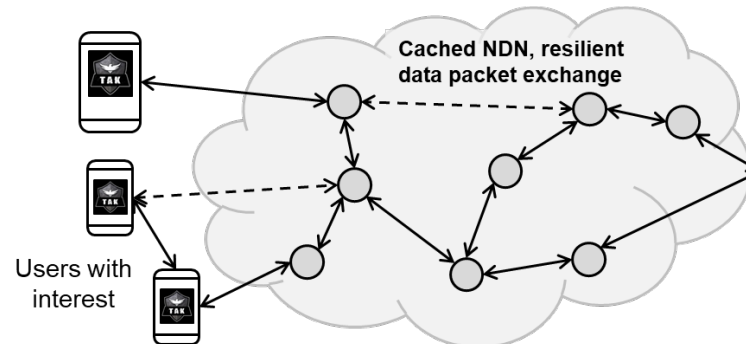


- Centralized, enterprise level approach has no way to establish trust at the tactical edge
- VPNs used to secure networks and devices quickly fail in high error rate tactical networks
- Configuring and deploying networks for US and coalition use takes months to years

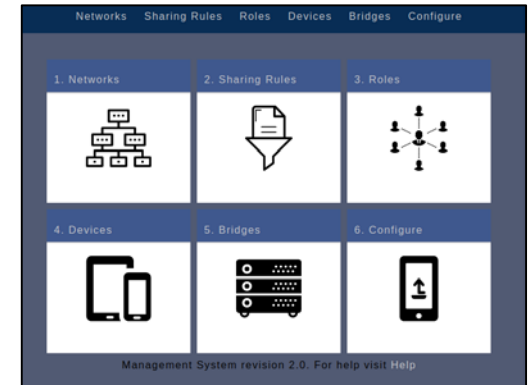
SHARE



Multiple levels of security on a single, trusted handheld



Resilient networking with built-in encryption, caching, and key distribution

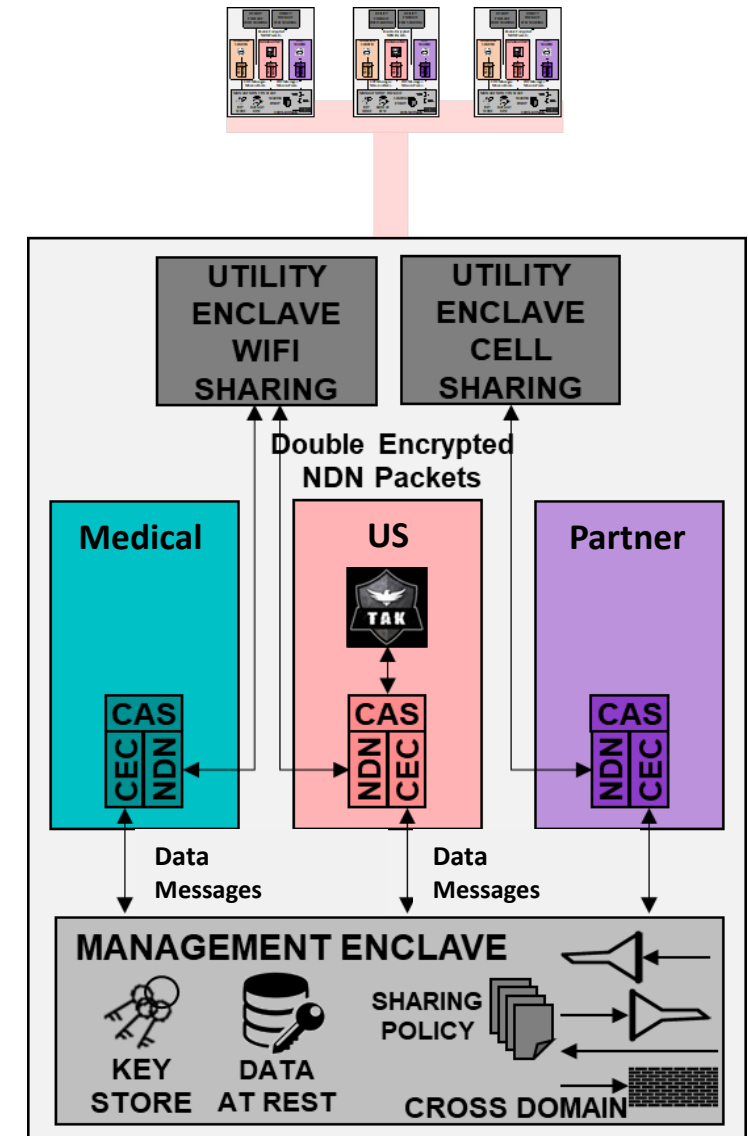


Software to automate and drastically reduce configuration time

Secure data sharing at multiple security levels between coalitions across mobile tactical networks

Secure Enclaves

- Multiple secure enclaves on a single phone
 - All data packets are doubly-encrypted to prevent leakage
- Strong isolation within CPU & power constraints of EUD
 - Enclaves contain full Android stacks with little-to-no impact on battery life
 - Access control reinforced through NSA compliant Security-Enhanced (SE) Linux policies & private crypto-backed data storage
 - Shared resources such as cameras and microphones are multiplexed, thus only visible to a single enclave at a time
- Least-privilege enclave management isolates key management, policies, and auditing from mission processing
- Design enables use of Information Centric Networking concepts such as NDN without impacting existing application(s), e.g. TAK

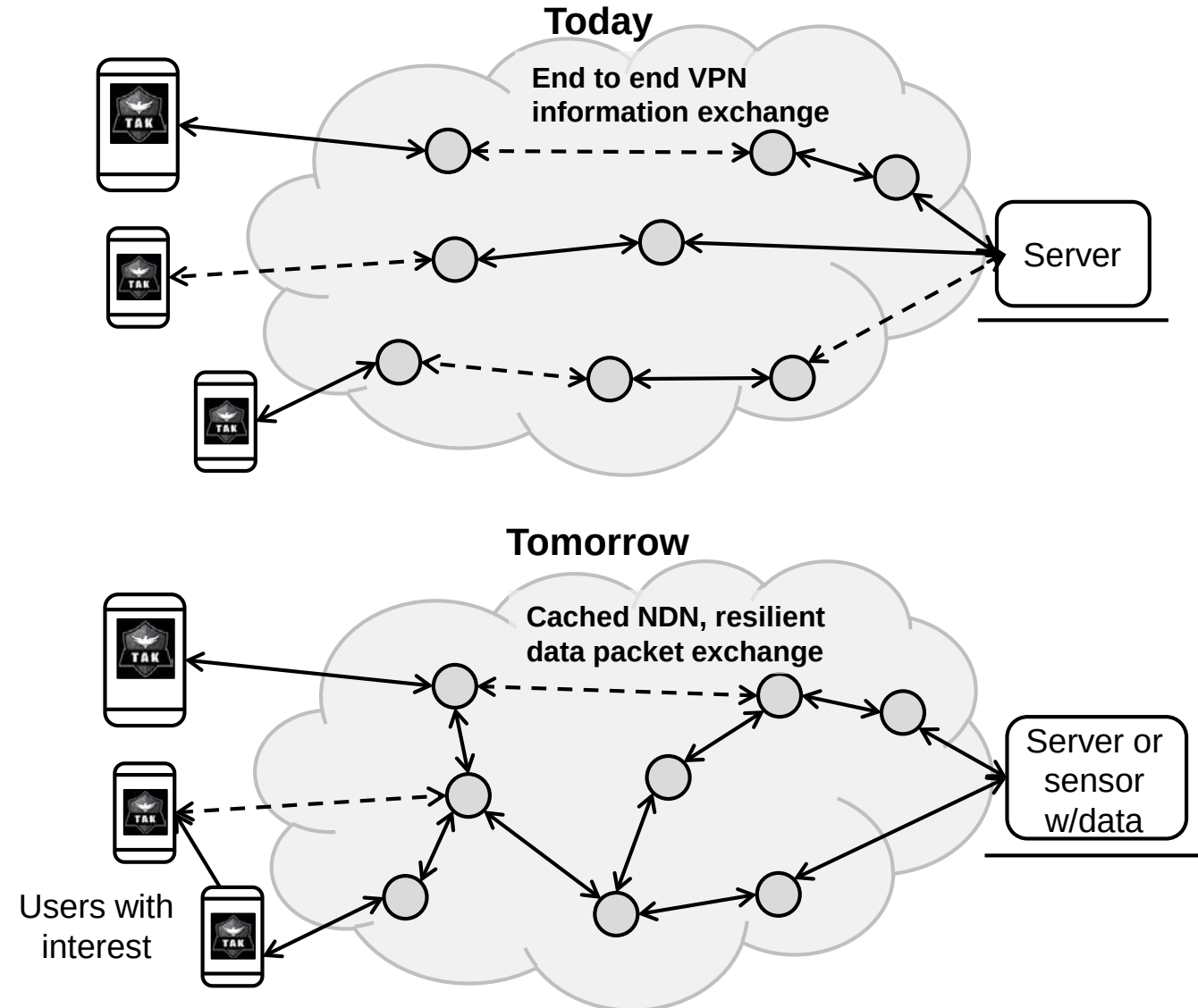


NSA: National Security Agency
EUD: End User Device
CPU: Central Processing Unit
NDN: Named Data Networking
TAK: Tactical Assault Kit

Resilient Secure Networks

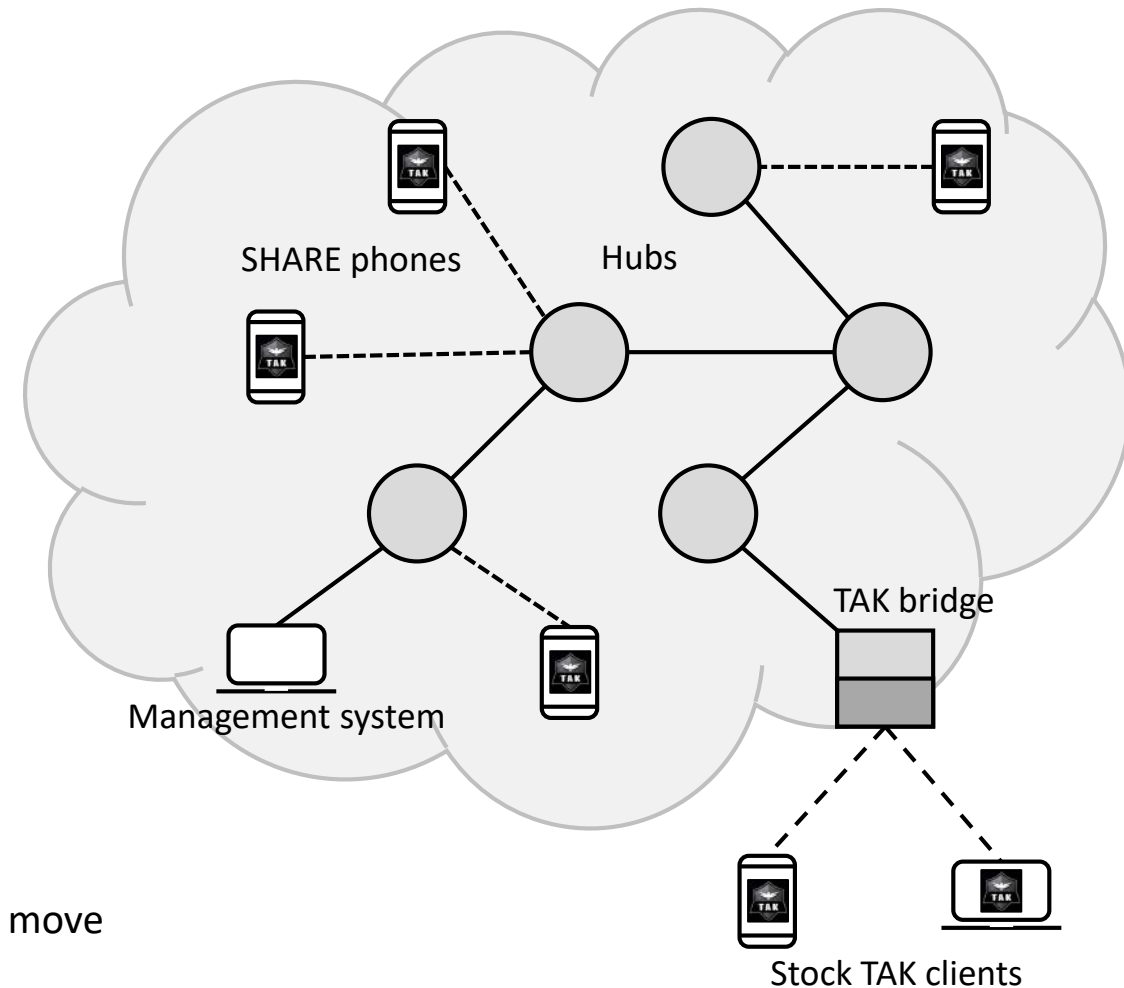
- Primary features of tactically hardened and secured Named Data networking (NDN) include:
 - Resilient and efficient packet forwarding to address mobility, i.e. links may be unstable but data packets, thus information flow, persist in the network
 - Rapid information recovery via data synchronization tailored to tactical network traffic patterns
 - Dual layer packet level security implementation that aligns with IPsec VPN data protections
 - Secure bridging of SHARE and non-SHARE networks and devices via NDN

Provides **security at the packet level** as each is authenticated, encrypted, and can securely transit the network on its own without a persistent, end-to-end connection



Adaptation of NDN for Tactical Use

- Network consists of smartphones, intermediate hubs to route encrypted data, and bridge nodes to extend comms
- Intermediate hubs include NDN packet forwarder, authentication daemon, and CRL management
- TAK bridge – connecting SHARE and non-SHARE networks
 - Houses TAK server and SHARE abstraction services to aggregate comms for a trusted group (e.g., partner forces)
 - Supports clients running stock TAK to display data for the trusted group
- Data synchronization tailored for tactical comms, providing various quality of service guarantees
 - ICT-Sync – secure synchronization protocol that is 100x more responsive than existing NDN sync protocols
 - Presence channel to allow users to keep track of contact lists, provides only most recent location update
 - Dedicated channels for comms within trusted group, e.g., group chats and sharing mission packs
 - Individual point-to-point channels for direct messaging
- Stateful, efficient routing strategy that tracks smartphones as they move throughout network

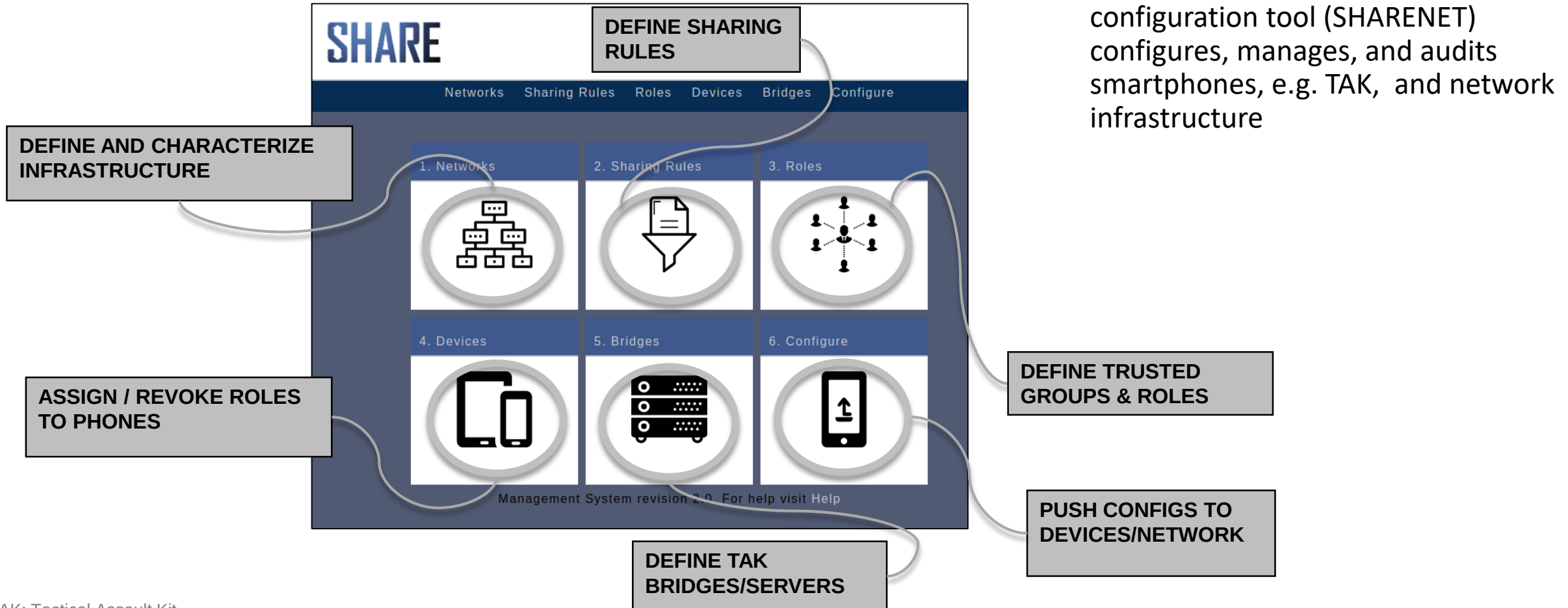


Making NDN Work for Tactical Comms

- Improvements span a variety of areas, including synchronization, scalability, routing, and topology design
- Migration from ChronoSync to ICT-Sync
 - ChronoSync wasn't well suited for tactical comms where multiple users are creating data concurrently
- Improved efficiency of sync protocol
 - Sync-to-data namespace convention removed the need for participant discovery; enables response-less sync traffic
- Mapping of information to sync instances and namespaces in each enclave
 - Dedicated presence channel used to discover new contacts and broadcast PLI updates
 - Broadcast channel used for enclave-wide chat messages and control messages
 - Individual channels for direct messages and group chats
- Addition of a scalable routing protocol (bestcast) that supports mobility of end users
 - Requests for data packets from individual users are properly routed as the end moves throughout the network
- Tiered topology for NDN overlay network
 - Each broadcast packet needs to be sent out over all next-hops; tiered approach reduces burden of routing sync traffic
- Multiple NFDs at each subnet
 - Single-threaded NFD implementation is potential choke point; multiple NFDs enabled load balancing of phone next-hops

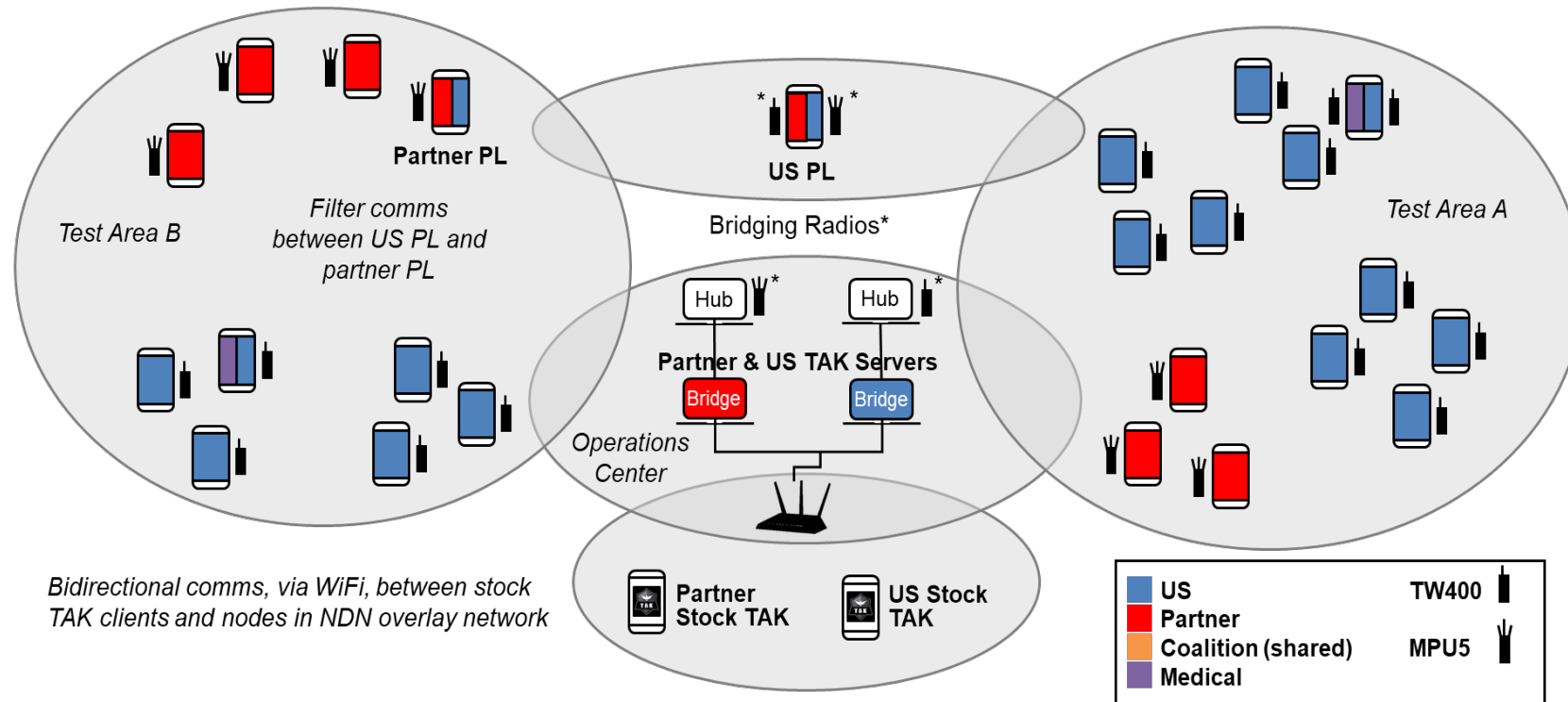
Configuration Management

SHARENET User Interface



TAK: Tactical Assault Kit

Phase 3 Test and Evaluation Field Event

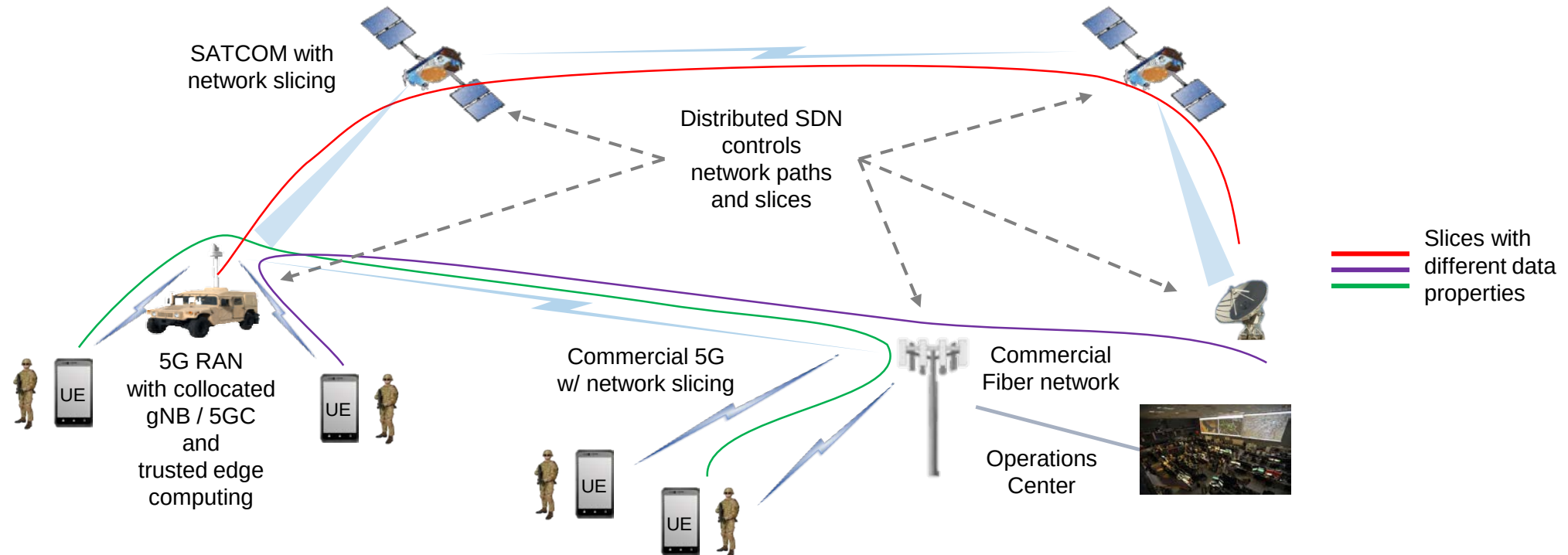


- Successfully conducted over the air test event from 27-30 July 2020
- With TAK software on commercial handsets demonstrated:
 - Multiple secure enclaves on handheld devices
 - Packet-level network security using NDN protocols
 - Persistent data sharing and synchronization
- Demonstrated resilience of NDN vs non-NDN network

TAK: Tactical Assault Kit
NDN: Named Data Networking
PL: Platoon Leader

5G Secure Networking with SHARE

- Joint effort between OUSD(R&E), DARPA, and NSF
- 5G cellular network technology poised to support critical military applications
- Goal: Demonstrate the secure exchange of information over 5G commercial and military networks employing network slicing to further isolate data



SDN: Software Defined Network
RAN: Radio Access Network

gNB: Next generation NodeB
5GC: 5G Core

OUSD(R&E): Office of the Undersecretary of Defense for Research and Engineering
NSF: National Science Foundation

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SHARE Transition

- Promoting SHARE technology within user communities to demonstrate solutions to requirements, interoperability, and drive adoption
 - Actively engaged with the 75th Ranger Regiment and the Telemedicine and Advanced Technology Research Center on transition efforts
- Working with NIST to standardize NDN to ensure compatibility across Government, industry, and academia
 - Interested in exploring TAK over NDN for public safety use cases
- Discussions with commercial partners to include multi-enclave and NDN technology in future versions of their phones